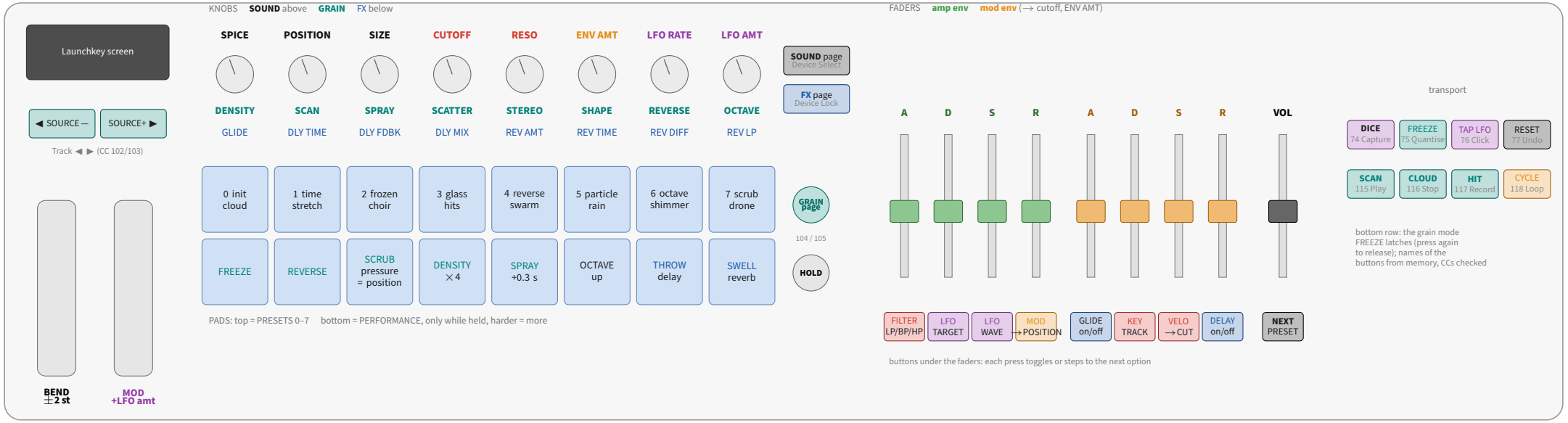
grains

Axoloti Core + Launchkey 49 MK3 (standalone mode) · mono granular synth, the SD-card patch · `patches/grains/grains_v1.axp`



Knobs: three pages, scaled pickup

SPICE	chaos: each phrase gets its own position (±0.25), pitch (±0.5 st) and cutoff (±1.5 oct) offset, and all grains more spray (+0.1 s), scatter (+3 st), reversal (+0.3) and width (+0.5)
POSITION	where in the source, 0 – 1 (SCAN: where a phrase starts)
SIZE	grain length 5 ms – 1 s (exponential)
knobs 4–8	CUTOFF, RESO, ENV AMT, LFO RATE, LFO AMT: as freak
DENSITY	0.5 – 200 grains/s; in HIT, grains per note, 1 – 20
SCAN	SCAN speed: centre = still, right forward, left backward, $\frac{1}{16}$ – 4×
SPRAY	random start, ±0 – 0.5 s
SCATTER	random pitch, ±0 – 12 st
STEREO	random pan width, 0 – 1
SHAPE	grain window: smooth (bell) → percussive (4 % attack)
REVERSE	chance a grain plays backwards
OCTAVE	chance a grain jumps an octave, up or down
FX page	as freak: GLIDE; delay (now stereo); reverb

Pages: Device Select = SOUND, round button (upper, right of the pads) = GRAIN, Device Lock = FX. A knob left behind by a page change, preset or DICE moves the value proportionally until they meet: no jumps.

Grain modes (Play / Stop / Record)

SCAN: grains at a steady rate from a play head that moves through the source at SCAN speed, from POSITION on each new phrase: time stretch, slow motion, reverse, still.

CLOUD: grains at random times (mean = DENSITY) around POSITION: textures.

HIT: each note fires DENSITY grains at once, then nothing: percussive. Short SHAPE and a plucky amp envelope suit it.

C4 (middle C) plays a source at its own pitch; keys transpose the grains (−48 to +36 st).

Keys: mono, legato, hold

The newest key sounds; release it and the previous one comes back, without restarting (legato). A key with no other down starts a *phrase*: envelopes restart, SCAN jumps to POSITION, SPICE rolls. GLIDE as freak. **HOLD** (round button, lower) or the sustain pedal keeps the note after release; a new key then starts a new phrase.

Envelopes, filter, LFO

As freak: faders 1–4 amp, 5–8 mod (1 ms – 10 s; S 0 – 1); CYCLE (Loop) loops the mod env. Mod env → cutoff (ENV AMT) and → position (+0.5) with MOD→POSITION. Grains keep coming while the amp envelope is open, release included. Filter: stereo 12 dB/oct SVF, LP/BP/HP. LFO TARGET: **position** (±0.5) → size (× $\frac{1}{4}$–4) → cutoff (±3 oct) → pitch (±2 st); WAVE and TAP as freak.

Sources

SOURCE (Track ◀ / ▶) steps through what is loaded: 0 = built-in (a saw chord that brightens along its 1.5 s), 1–8 = the starter bank (page 2), then your files. Presets name a source; one that is not loaded plays the built-in one.

Your own: WAV files in `/grains` on the SD card, up to 16, loaded in name order (name them `09_xxx.wav`, at most 8 characters before `.wav`). 16, 24 or 32-bit or float; any rate; stereo is mixed to mono. Memory: 7.2 MB in all, 3 MB of it the starter bank, so about 45 s at 48 kHz for your own (the log shows what is left). Put them there with *Board → Enter Card Reader Mode*.

Loading runs in the background after power-up (the starter bank: 1.1 s): until a source is there, the built-in one plays.

Pads

Top = presets 0–7. **Bottom** = performance, while held; strike velocity, then pressure, sets the amount:

FREEZE	the position stops (SCAN head too)
REVERSE	grains backwards, SCAN runs backwards
SCRUB	pad pressure *is* the position
DENSITY / SPRAY	up to 4× / +0.3 s
OCTAVE	+12 st
THROW / SWELL	delay throw / reverb swell, as freak

Two patches, one board

SD card in at power-up: grains starts. **No card**: freak (in flash) starts.

Install: open `grains_v1.axp`, *Upload to SD Card as Startup*. That also copies the starter bank to `/grains` (and replaces an older startup patch on the card).

If grains will not start: power off, take the card out: freak starts. Then see page 2.

Signal path and load

20 grains (pool) → stereo SVF → amp env × velocity → stereo delay → Elements reverb → VOL. When the pool is full, new grains wait: the load stays bounded. DSP 41 % idle, 75.5 % with all 20 grains; patch RAM 28 of 44 KB.

Notes

Toggle states are not shown (no LEDs in standalone mode). With the patcher connected, its log reports the loaded files, pages, presets, modes, sources and knobs catching up.

Starter bank: `/grains/01..08.wav`, 4 s each, made with freak's oscillators at C4

(each changes along its length, so POSITION matters)

1	vowel FOF on C and G: a → e → i → o → u along the source
2	paraphonic chords, a new chord each second
3	harmonics, their centre rising (dark → bright)
4	bell, 8 strikes (one per 0.5 s), more inharmonic each time
5	particles, density and pitch spread rising
6	feedback FM, more modulation along it, a new ratio each second
7	plucked arpeggio C E G B D B G E, 8 notes a second
8	saw swarm C3 G3 C4, detune rising, filter opening

Presets: top pads = 0–7; NEXT PRESET (fader button 9) steps; RESET (Undo) = 0

(chosen from the parameter meanings, not yet tuned by ear)

0 init cloud	CLOUD on source 1 (vowels), middle; 140 ms grains, 22/s, half width; LP 4.7 kHz; delay off. The power-on sound.
1 time stretch	SCAN source 2 (chords) from the start at half speed: every chord lasts twice as long, pitch unchanged.
2 frozen choir	SCAN still, source 1 at 0.43: 320 ms grains, 44/s, wide, some octaves; slow triangle LFO drifts the position; attack 0.33 s, release 3 s. Hold a note.
3 glass hits	HIT on source 4 (bells) at a strike: 11 percussive grains per note, wide, some octaves; decay 0.7 s. Play rhythms.
4 reverse swarm	CLOUD on source 8 (saw swarm): 87 % of the grains backwards, S&H LFO on grain size.
5 particle rain	CLOUD on source 5 (particles): 40 ms grains at 90/s, spray 0.25 s, ±2.7 st, full width, percussive; S&H LFO on cutoff.
6 octave shimmer	CLOUD on source 3 (harmonics): long grains, 80 % jump an octave; attack 0.7 s, release 3 s; big reverb.
7 scrub drone	SCAN source 7 (plucks) at 0.07×: a frozen arpeggio slowly turning; CYCLE on; glide.

A preset sets the mode, the source, all three knob pages, faders 1–8 and the toggles; the volume stays. **DICE** (Capture MIDI) rolls the mode and the SOUND and GRAIN pages instead; source, envelopes, FX and volume stay.

Sound design moves

Time freeze: SCAN with SCAN at the centre, or hold FREEZE; then sweep POSITION by hand or SCRUB with pressure. **Texture from anything**: CLOUD, SIZE 1/2, DENSITY 3/4, SPRAY 1/4, STEREO full. **Glitch**: HIT, SIZE 1/4, SHAPE full, SCATTER 1/2, play fast. **Moving drone**: LFO TARGET position, slow triangle, MOD→POSITION with CYCLE on. **Pitched clouds**: SCATTER 0, OCTAVE 1/2, the Launchkey's Scale mode.

Where the values fall (knob turned . . .)

	1/4	1/2	3/4	full
SIZE	19 ms	72 ms	0.26 s	1 s
DENSITY (grains/s)	2.3	10	44	200
DENSITY in HIT (grains/note)	6	11	15	20
SCAN (× real time)	−0.45	still	+0.42	+4
SPRAY (±)	32 ms	0.13 s	0.28 s	0.5 s
SCATTER (±)	0.8 st	3 st	6.7 st	12 st

SCAN: still from 61 to 67 (the centre); ×1 at about 5/6 of the travel (and −1 at 1/6). Envelopes, CUTOFF, RESO, ENV AMT, LFO, GLIDE, delay and VOL fall as on the freak card.

Something seems wrong?

No sound: VOL up? In HIT, sound only comes on a key press. SIZE and DENSITY not both at the bottom? CUTOFF as on freak.

Only the built-in sound: no `/grains` folder, or the files are still loading or not WAV (the log says which). Upload the patch to the SD card again to copy the bank.

Sound stuck on: HOLD or the sustain pedal. Press HOLD (round button, lower).

Nothing moves: FREEZE latched (Quantise), or SCAN at its centre.

grains does not start, or hangs: power off, remove the card, power on: freak starts from flash. Fix the card with Card Reader Mode (rename `/start.bin` to skip grains).

Lost: RESET (Undo).

MIDI reference (beyond the freak card)

GRAIN page / HOLD (round buttons)	ch 16 CC 104 / 105
SCAN, CLOUD, HIT (Play, Stop, Record)	ch 16 CC 115, 116, 117
FREEZE latch (Quantise)	ch 16 CC 75
SOURCE − / + (track ◀/▶)	ch 16 CC 102 / 103
sustain pedal (= HOLD while down)	ch 1 CC 64